

Dr. Fares AL-Suwailh

Exercises 101

Exercises ①

(101)

Dr. Fares

$$3.3 \quad ① \quad \lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$$

Ans: 0

$$② \quad \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{|x|} \right)$$

Ans: D.N.E

$$③ \quad \lim_{x \rightarrow 3} (x-3) \sqrt{1 + \frac{1}{(x-3)^2}}$$

Ans: D.N.E

$$④ \quad \lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x-1}^2}$$

Ans: D.N.E

$$5.3 \quad ⑤ \quad \lim_{x \rightarrow \infty} \left[\sqrt{x} \sin\left(\frac{1}{x}\right) \right]$$

Ans: 0

$$⑥ \quad \lim_{x \rightarrow 4} \left(\frac{1}{x-4} - \frac{3}{x^2-5x+4} \right)$$

Ans: $\frac{1}{3}$

$$⑦ \quad \lim_{x \rightarrow 2} \frac{\sqrt{4-x} - \sqrt{x}}{x-2}$$

Ans: $-\frac{1}{\sqrt{2}}$

$$⑧ \quad \lim_{x \rightarrow 2} (x^2 - 4x + 4) \cos\left(\frac{2}{x-2}\right)$$

Ans: 0

$$3.3 \quad ⑨ \quad \lim_{x \rightarrow 0} \frac{2x + \sin 3x}{x - \tan 6x}$$

Ans: -1

$$⑩ \quad \lim_{t \rightarrow 1} \frac{t^4 - 1}{t^3 - 1}$$

Ans: $\frac{4}{3}$

$$⑪ \quad \text{Show that } \lim_{x \rightarrow 3} \lfloor x \rfloor \text{ D.N.E}$$

$$⑮ \quad \lim_{x \rightarrow 0^+} \frac{\ln(4/5)}{x}$$

Ans: $-\infty$

⑫ Solve: Ques: 51, 57, and 58 Page 108

$$⑬ \quad \lim_{x \rightarrow \infty} \frac{\ln(4x)}{\ln(5x)}$$

→ Ans: ①

$$⑭ \quad \lim_{x \rightarrow 0^+} \frac{\ln(4x)}{\ln(5x)}$$

→ Ans: $\frac{1}{1}$

Dr. Fares AL-Sawalh

Exercises 2 (10)

① Find and Classify (R.D, J.D, I.D) the discontinuities of

↙ a) $f(x) = \begin{cases} \frac{x-5}{x^2-6x+5} & x < 0 \\ \frac{x^2+x}{x^3-x} & 0 < x < 1 \\ \frac{-2|x-2|}{x^2-3x+2} & x > 1 \end{cases}$

Ans

$x=0$	R.D
$x=1$	I.D
$x=2$	J.D

↙ b) i) $f(x) = \begin{cases} x^2 - \sqrt{8-x} & x < -1 \\ -2 & x = -1 \\ \frac{x^2-3x+2}{x|x-2|} & x > -1 \end{cases}$

Ans

$x=0$	I.D
$x=2$	J.D
$x=-1$	Conf.

ii) on what intervals is the function f continuous?

Ans: $\mathbb{R} \setminus \{0, 2\}$

↙ c) $f(x) = \begin{cases} \frac{x^2-x}{x^3} & x < -1 \\ 2 & -1 \leq x < 1 \\ 3 & x=1 \\ \frac{x+1}{x^3-1} & 1 < x < 2 \\ \frac{x^2-1}{x^2-3x+2} & x > 2 \end{cases}$

Ans

$x=-1$	J.D
$x=1$	R.D
$x=2$	I.D

↙ d) $f(x) = \begin{cases} \frac{x^2-4}{(x+2)(\sqrt{2-x})} & x \leq -1 \\ \frac{1}{\sqrt{x^2-4x+4}} & x > -1 \end{cases}$

Ans

$x=-2$	R.D
$x=-1$	J.D
$x=2$	I.D

Ans

$x=1$	J.D
$x=2$	R.D
$x=-2$	I.D

↙ e) $f(x) = \frac{(x-2)|x-1|}{(x-1)(x^2-4)}$

⑧ Find nonzero values for the constant A and B, if any, that make the function f cont. at $x=1$, where

$$f(x) = \begin{cases} \frac{A(x-1)}{x^2-5x+4} & x < 1 \\ \frac{B}{4} & x = 1 \end{cases}$$

$$f(x) = \begin{cases} \frac{A(x-1)}{x^2-5x+4} & x < 1 \\ \frac{B}{4} & x > 1 \end{cases}$$

Ans: $A=3$
 $B=4$

② Find the limit of the following

(a) $\lim_{x \rightarrow \infty} \frac{x \sin x}{x^2+1}$

Ans: 0

(b) $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - x)$

Ans: $\frac{1}{2}$

(c) $\lim_{x \rightarrow \infty} (\sqrt{4x^2-1} - 2x)$

Ans: 0

(d) $\lim_{x \rightarrow -\infty} x + \sqrt{x^2+3x+1}$

Ans: $-\frac{3}{2}$

(e) $\lim_{x \rightarrow \pi} (x-\pi)^2 \cos\left(\frac{1}{\pi-x}\right)$

Ans: 0

(f) $\lim_{x \rightarrow 0} \frac{x}{2+\sin(\frac{1}{x})}$

Ans: 0

g) $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{|x-1|}}$

Ans: DNE

h) $\lim_{x \rightarrow \infty} (x^2 - x + 1)$

Ans: ∞

* Suppose f and g are cont. functions s.t:

$g(2) = 6$ and $\lim_{x \rightarrow 2} [3f(x) + f(x)g(x)] = 36$. Find $f(2)$

Ans: 4

④ Find the (V.A) and (H.A) (if any)

(a) $f(x) = \frac{\sqrt{x^2+1}}{x-3}$

(c) $f(x) = \tan x$

Ans: $x = \pm \frac{\pi}{2}$
V.A.
No H.A.

(b) $f(x) = \frac{|x|(2x^2+3)}{x^3+8}$

Ans: $x = -2$ V.A.
 $y = \pm 2$ H.A.

(d) $f(x) = \ln x$

$x=0$ V.A.
No H.A.

⑤ Show that f is cont. on $[-3, 3]$ if $f(x) = \frac{\sqrt{9-x^2}}{x^2-16}$

⑥ Use I.V.T. to show that $e^x + x = 0$ has at least one real root

⑦ let $f(x) = x - \ln(x^3)$. Use I.V.T to show that f has at least one real root on $[1, e^7]$.

Section 3.9 : Related Rates

- 1) Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{second}$. How fast is the radius of the balloon changing when the diameter is 50 cm?

$$V(\text{sphere}) = \frac{4}{3} \cdot \pi \cdot r^3$$

- 2) A ladder 10 feet long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 feet/second, how fast is the top of the ladder sliding down when the bottom of the ladder is 6 feet from the wall?

- 3) Car A is traveling west at 30 mile/hour and car B is traveling north at 40 mile/hour. Both started from the same point. At what rate the distance between them is changing after 2 hours.

- H · W 4) Car A is traveling west at 50 mile/hour and car B is traveling north at 60 mile/hour. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mile and car B is 0.4 mile from the intersection.

- 5) A water tank has the shape of an inverted circular cone with base radius 2 meters and height 4 meters. If water is being pumped into the tank at a rate of $2 \text{ meter}^3/\text{minute}$, find the rate at which the water level is rising when the water is 3 meters deep.

$$V(\text{right circular cone}) = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$$

3.10 Exercises : 4

① Estimate $\sqrt{35}$

② Approximate $\tan(44^\circ)$

③ Use differential to approximate
 $1 + (8.01)^{2/3}$

④ Let $f(x) = \frac{x^2}{2} - 3x^2 + 5$

① Find the linearization of $f(x)$ at $a=1$.

② Find linear approximation of $f(x)$ at $a=1$

⑤ ① Find the linearization of $f(x) = \sqrt{x+3}$ at $a=1$

② Use part ① to approximate $\sqrt{3.98}$

⑥ Let $f(x) = \sin^2(x^2 - 3x + \frac{\pi}{4})$

① Find $f'(x)$

② Use linear approximation to estimate $f(0.02)$

⑦ Find the differential of each function:

① $y = e^{\tan(\pi t)}$

② $y = \sqrt{1 + \ln x}$

⑧ Use differential to approximate

① $\sqrt[3]{1001}$

② $\frac{1}{4.002}$

Dr. Fares

4.2 The Rolle's Theorem and The Mean Value Theorems

H.W. $\textcircled{1}$ Use the Mean Value Theorem to show that
discuss

$$(3+x)^{1/3} < 2 + \frac{1}{2}(x-5) \text{ for all } x > 5.$$

2. Let $f(x) = \frac{x+1}{x-1}$.

(a) Show that there is no real number $c \in (0, 2)$ such that $f(2) - f(0) = 2f'(c)$.

(b) Why this does not contradict the Mean Value Theorem?

H.W. $\textcircled{3}$ Show that $f(x) = x^4 + 2x^2 - 3x + 1$ has exactly one critical number.
discuss

4. Let

$$f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1, \\ 2-x & \text{if } 1 < x \leq 2. \end{cases}$$

(a) show that $f(0) = f(2)$,

(b) show that $f'(c) \neq 0$ for all $c \in (0, 2)$,

(c) Does this contradict the Rolle's Theorem? Explain.

5. Suppose that f is a continuous function on $[a, b]$ and $f'(x) < 0$ for all $x \in (a, b)$.
Use the Mean Value Theorem to show that $f(b) < f(a)$.

6. Let f, f' , and f'' be continuous functions on $[a, b]$. Suppose that the graph of f intersect the x -axis at three points in (a, b) . Show that $f'' = 0$ has a solution in (a, b) .

H.W. $\textcircled{7}$ Let $f(x) = 3x + \frac{9}{5}(x-1)^{5/3} + 1$. Show that there is no $c \in (0, 2)$ such that $f''(c) = 0$.
Explain why this result does not contradict the Rolle's Theorem.

⑧ Use the M.V.T, to show that for $x > 0$
 $\tan^{-1} x < x$.

⑨ Use the M.V.T, to show that
 $\tan\left(\frac{3}{2}\right) - \tan\left(\frac{1}{2}\right) > 1$

⑩ Use the M.V.T to show that
 $|\cos^7 u - \cos^7 v| \leq 7 |u - v|$.

⑪ Let $f(x) = x^5 + 6x^3 - \sqrt{2}$

Use Rolle's Th. to show that

f cannot have more than one inflection point
on $[0, 1]$.

⑫ Suppose that f is a diff. on $\mathbb{R}$
satisfying the following condition

$$f(-x) = x^2 f(x).$$

Use Rolle's Th. to show that $\exists c \in (-1, 1)$ s.t.
 $f'(c) = 0$.

* Increasing / Decreasing Test:

- a) If $f'(x) > 0$ on an interval I , then f is an increasing ($\nearrow$) function on I .
- b) If $f'(x) < 0$ on an interval I , then f is an decreasing ($\searrow$) function on I .

* The First Derivative Test:

Suppose that c is a critical number of a continuous function f .

- a) If f' changes from $+$ to $-$ at c , then f has a local maximum at c .
- b) If f' changes from $-$ to $+$ at c , then f has a local minimum at c .
- c) If f' does not change sign at c , then f has no local extrema at c .

* Concavity Test:

- a) If $f''(x) > 0$ for all x in interval I , then the graph of f is concave upward (CU) on I .
- b) If $f''(x) < 0$ for all x in interval I , then the graph of f is concave downward (CD) on I .

* The Second Derivative Test:

Suppose that f'' is continuous near a number c .

- a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

* Curve Sketching Guidelines:

1. Domain: find the domain of the function f , denoted by D_f .
2. intercepts: find y -intercept $(0, f(0))$ and x -intercept $(x, 0)$.
3. Symmetry:
 - a) if $f(-x) = f(x)$ for all $x \in D_f$, then f is even function and the curve is symmetric about the y -axis. That is, $(a, b) \mapsto (-a, b)$. Take f to be x^2 , x^4 , or $\cos(x)$ as an example.
 - b) if $f(-x) = -f(x)$ for all $x \in D_f$, then f is odd function and the curve is symmetric about the origin. That is, $(a, b) \mapsto (-a, -b)$. Take f to be x , x^3 , or $\sin(x)$ as an example.
4. Asymptotes: Horizontal asymptotes (H.A.) and Vertical asymptotes (V.A.).
5. Increasing/Decreasing Test: interval on which f is increasing or decreasing.
6. Local extrema: find all max. and min. local extrema.
7. Concavity and Inflection Points: find where the curve of f is CU or CD and find all of the inflection points.
8. Sketch: use all of the previous information to sketch the curve of f .

① ←

إذا احتجنا لهم !
أو طلبهم مراحلة
بالإختبار

② ←

③ ← الجدول الأول

④ ←

الجدول الثاني

⑤ ←

(*) 1st derivative test:

$f(x) \rightarrow$ given

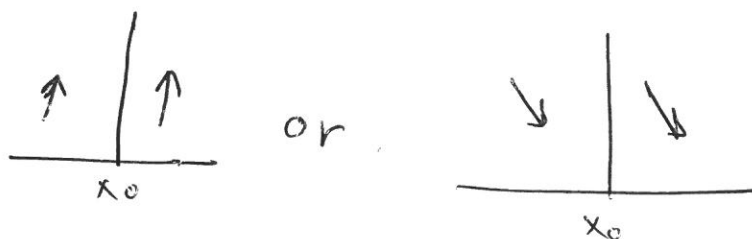
- ① D_f
 ② $C.N$ $\begin{cases} \rightarrow f'(x) = 0 \\ \rightarrow f'(x) \text{ D.N.E} \end{cases}$

$\forall x_0 \in \underline{\underline{D_f}} \quad (x_0 \in C.N)$

③

D_f	
f'	
$\rightarrow \downarrow$	

Note that:



No local max. or min



loc. max at x_0
 Point $(x_0, f(x_0))$

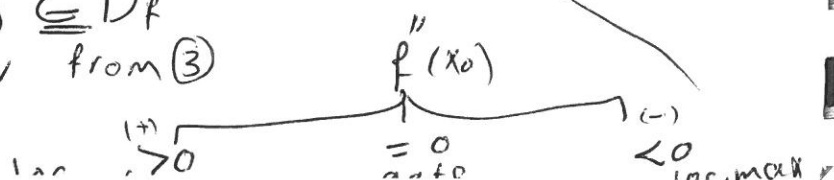
or



loc. min at x_0
 Point $(x_0, f(x_0))$

(*) 2nd der. test:
 $f(x) \rightarrow$ given

- 1) D_f
- 2) $f'(x), f''(x)$
- 3) $C.N \quad (x_1, x_2, \dots) \in \underline{\underline{D_f}}$
- 4) $\forall x_0 \in D_f$ is C.N from ③



Exercises from 3.1 $\rightarrow$ 4.2

① Let $f(x) = a^x$. Use the def. of the derivative to find $f'(x)$. [Hint: see the book P179].

② $\lim_{x \rightarrow 0} \frac{3^x - 1}{x}$

③ $\lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{3(x^2 - 2x + 1)}$

④ Find $\frac{dy}{dx}$ where

① $y = (\sin x)^{\ln x}$

② $y = \ln(\ln(\ln x))$

③ $y = \left((\ln x)^{\ln x} \right)^{\ln x}$

⑤ Let $f(x) = e^x$

① Find the linearization of f at $a = 0$.

② Use linear approximation to estimate $e^{0.01}$

⑥ Find y' if $y = \cos\left(\frac{1 - e^{2x}}{1 + e^{2x}}\right)$

⑦ Find y' if $y = \sin(\sin(\sin(x)))$

⑧ Find y' if $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$

(2)

9) Find y' if:

(a) $y = 2^{3x^2}$

(b) $\sin^2(e^{\sin^2 t})$

(c) $\cos \sqrt{\sin(\tan \pi x)}$

10) Find $\frac{dy}{dx}$

(a) $\tan^{-1}(x^2 y) = x + xy^2$

(b) $e^y \cos x = 1 + \sin(xy)$

(11) If $xy + e^y = e$, find the value of y' at the point where $x=0$.(12) If $f(x) + x^2 [f(x)]^3 = 10$ and $f(1) = 2$, find $f'(1)$ (13) Differentiate f and find the domain of f

(a) $f(x) = \frac{x}{1 - \ln(x-1)}$

(b) $f(x) = \ln(x^2 - 2x)$

(c) $y = (2x+1)^5 (x^4-3)^6$

(d) $y = \sqrt{\frac{x-1}{x^4+1}}$

(e) Use the def. of derivative to prove that

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

(f) show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \forall x > 0$$

(14) Example (5) P247

(15) A rocket, rising vertically is tracked by a radar station that is on the ground 4 km from the launchpad. How fast is the rocket rising when it is 3 km high and its distance from the radar station is increasing at a rate of 1500 km/h?

(16) A particle is moving along a hyperbola $xy=8$. As it reaches the point (4,2), the y-coordinate is decreasing at a rate of 3 cm/s. How fast is the x-coordinate of the point changing at that instant?

(17) (a) Find the linearization of the function $f(x) = \sqrt{x+3}$ at $a=1$

(b) Use part (a) to approximate $\sqrt{3.98}$

(18) Approximate (a) $\tan 44^\circ$

(b) $\sec(0.08)$

(c) $\ln(1.05)$

مطلوب
(19)

Find $f'(x)$ if it is known that $\frac{d}{dx} [f(2x)] = x^2$.

[20] Example (P275) : $3.1 \rightarrow 4.2$

[21] If a and b are positive numbers, find the maximum value of $f(x) = x^a(1-x)^b$, $0 \leq x \leq 1$

[22] Find the absolute extreme values of f :

(a) $f(t) = 2 \cos t + \sin 2t$, $[0, \frac{\pi}{2}]$

(b) $f(x) = x e^{-\frac{x^2}{8}}$ $[-1, 4]$

(c) $f(x) = \ln(x^2 + x + 1)$ $[-1, 1]$

[d] $f(x) = x \sqrt{x - x^2}$

(23) Show that the equation $x^3 - 15x + C = 0$ has at most one root in the interval $[-2, 2]$

(24) Show that a poly. of degree 3 has at most 3 real roots.

(25) If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$ how small can $f(4)$ possibly be?

(26) Does there exist a function f s.t.: $f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$, $\forall x$

(27) Use the MVT to prove the inequality $|\sin a - \sin b| \leq |a - b|$ $\forall a$ and b

(28) Let $f(x) = \frac{1}{x}$ and $g(x) = \begin{cases} \frac{1}{x} & x > 0 \\ 1 + \frac{1}{x} & x < 0 \end{cases}$ Show that $f'(x) = g'(x) \forall x \in \mathbb{D}$.
(b) Show that f is a constant.

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Exercises from 4.3 and 4.5 and 4.7

(1) Find the local max. and min. values of the function $g(x) = x + 2\sin x$ $0 \leq x \leq 2\pi$

(2) Sketch a possible graph of a function f that satisfies the following conditions:

(i) $f'(x) > 0$ on $(-\infty, 1)$, $f'(x) < 0$ on $(1, \infty)$

(ii) $f''(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$
 $f''(x) < 0$ on $(-2, 2)$

(iii) $\lim_{x \rightarrow -\infty} f(x) = -2$, $\lim_{x \rightarrow \infty} f(x) = 0$

[Hint: See P294]

(3) sketch the graph of $f(x) = e^{\frac{1}{x}}$

(4) $\simeq \simeq \simeq \simeq f(x) = x^{\frac{2}{3}}(6-x)^{\frac{1}{3}}$

(5) $\simeq \simeq \simeq f(x) = \sqrt{x} - \sqrt[4]{x}$

(6) $\simeq \simeq \simeq f(x) = x^2 \ln x$

(7) $\simeq \simeq \simeq f(x) = x^2 - x - \ln x$

(8) $\simeq \simeq \simeq f(x) = \frac{x^2 - 4}{x^2 + 4}$

→

⑥

⑨ sketch the graph of $f(x) = 1 + \frac{1}{x} + \frac{1}{x^2}$

⑩ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \ln(1 - \ln x)$

⑪ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \frac{\cos x}{2 + \sin x} \quad \left[\begin{smallmatrix} \text{ser} \\ P_3 \end{smallmatrix} \right]$

⑫ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \ln(\sin x)$

⑬ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \frac{x - x^2}{2 - 3x + x^2}$

⑭ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \tan^{-1}\left(\frac{x-1}{x+1}\right)$

⑮ $\hookrightarrow \hookrightarrow \hookrightarrow \hookrightarrow f(x) = \sinh(x)$

Exercises for 4.7:

- ① Find the dimensions of a rectangle with perimeter 120m whose area is maximum
- ② Find the area of the largest rectangle that can be inscribed in a semi-circle of radius r .
- ③ Find the length and width of a rectangle whose perimeter is 16 and area is largest
- ④ Find the maximum value of P if $P = x^2 + y^2$ where $x - y = 40$
- ⑤ Find two +ve numbers s.t.: the sum of the first number and four times the second number is 80 and the product numbers is maximum.
- ⑥ A rectangular box with a square base and open top has a volume 4 m^3 . Find the dimensions of the box that minimize the surface area.

① Exercise number ① P38 and number ③ 8

② Exercise number ① P38

③ Exercise number 21, 23, and 25.

④ Exercise number ② and ③ P38

Bouns



⑤ $\int_{-1}^2 |x| dx$

⑥ $\int_0^{10} |x-5| dx$

⑦ $\int_{-\pi}^{\pi} \sin^2 x \cos^4 x dx$

show that

$$\int_1^3 \sqrt{x^4+1} dx \geq \frac{26}{3}$$

⑧ Given that $\int_0^1 3x \sqrt{x^2+4} dx = 5\sqrt{5} - 8$, what is $\int_1^0 3u \sqrt{u^2+4} du$?

⑨ write as a single integral in the form $\int_a^b f(x) dx$:

$$\int_{-2}^3 f(x) dx + \int_2^5 f(x) dx - \int_{-2}^{-1} f(x) dx$$

⑩ If $\int_0^9 f(x) dx = 37$ and $\int_0^9 g(x) dx = 16$, find

$$\int_0^2 [2f(x) + 3g(x)] dx$$

⑪ Use the properties of integrals to verify:

① $\int_0^4 (x^2 - 4x + 4) dx \geq 0$

② $\int_0^1 \sqrt{1+x^2} dx \leq \int_0^1 \sqrt{1+x} dx$

③ $2 \leq \int_{-1}^1 \sqrt{1+x^2} dx \leq 2\sqrt{2}$



(12) estimate:

(a) $\int_1^4 \sqrt{x} dx$

(b) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \tan x dx$

(c) $\int_0^2 x e^{-x} dx \rightarrow \text{Boukas}$ $\int_0^1 f(x) dx$

[d] (a) If f is cont. on $[a, b]$, Show that

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Hint: $-|f(x)| \leq f(x) \leq |f(x)|$.

(b) Use Part (a) to show that

$$\left| \int_0^{2\pi} f(x) \sin(2x) dx \right| \leq \int_0^{2\pi} |f(x)| dx$$

(13) What is wrong with the following calculation?

$$\int_{-1}^3 \frac{1}{x^2} dx = \left. \frac{x^{-1}}{-1} \right|_{-1}^3 = -\frac{1}{3} - 1 = -\frac{4}{3}$$

[read example (9) P 398]

[14] Find derivative of the following

(a) $g(x) = \int_1^x \frac{1}{t^3+1} dt$

(b) $h(x) = \int_1^{e^x} \ln t dt$

(c) $F(x) = \int_x^\pi \sqrt{1+\sec t} dt$

(d) $y = \int_0^{\tan x} \sqrt{t+\sqrt{t}} dt$

(e) $g(x) = \int_{2x}^{3x} \frac{u^2-1}{u^2+1} du$ [Hint: $\int_{2x}^{3x} f(u) du = \int_{2x}^0 f(u) du + \int_0^{3x} f(u) du$]

(f) $F(x) = \int_x^{x^2} e^{t^2} dt$

(g) $y = \int_{\cos x}^{\sin x} \ln(1+2u) du$

(15) on what interval is the curve

$$y = \int_0^x \frac{t^2}{t^2+t+2} dt \quad \text{concave downward?}$$

(16) IF $f(x) = \int_0^{\sin x} \sqrt{1+t^2} dt$ and $g(y) = \int_3^y f(x) dx$ find $g''(\frac{\pi}{6})$ (17) IF $f(1) = 12$, f' is cont. and

$$\int_1^4 f'(x) dx = 17, \text{ what is the value of } f(4)?$$

(18) (a) Show that $1 \leq \sqrt{1+x^3} \leq 1+x^3$ for $x \geq 0$.(b) Show that $1 \leq \int_0^1 \sqrt{1+x^3} dx \leq 1.25$

(19) Evaluate the following:

(a) $\int (x^2 + x^{-2}) dx$

(b) $\int_{-1}^1 (u+4)(2u+1) du$

(c) $\int_{-\pi/2}^{\pi/2} (1 + \tan^2 x) dx$

(d) $\int \sec t (\sec t + \tan t) dt$

(e) $\int_{-1}^2 ((x-2)|x|) dx$

(f) $\int_0^1 x \left(\sqrt[3]{x} + \sqrt[4]{x} \right) dx$

(g) $\int_{-10}^{10} \frac{2e^x}{\sinh x + \cosh x} dx$

(20) Evaluate: (a) $\int x^2 \sqrt{x^3+1} dx$

(b) $\int \cos^3 \theta \sin \theta d\theta$

(c) $\int \frac{dx}{5-3x}$

(d) $\int \frac{(\ln x)^2}{x} dx$

(e) $\int u \sqrt{1-u^2} du$

(f) $\int \sqrt{x} \sin(1+x^{3/2}) dx$

(g) $\int_{\pi/4}^{\pi/2} \sqrt{\cot x} \csc^2 x dx$

(21) Evaluate:

(a) $\int_1^2 \frac{e^{1/x}}{x^2} dx$

(b) $\int_0^1 (3t-1)^{50} dt$

(c) $\int_0^\pi \sec^2\left(\frac{t}{4}\right) dt$

(d) $\int_0^a x \sqrt{a^2 - x^2} dx$

(e) $\int_e^{e^4} \frac{dx}{x \sqrt{\ln x}}$

o-r

[22] Verify that $f(x) = \sin^3 \sqrt{x}$ is an odd function and use that fact to show that

$$0 \leq \int_{-2}^3 \sin^3 \sqrt{x} dx \leq 1$$

o-r

[23] If f is cont. and $\int_0^4 f(x) dx = 10$ find $\int_0^4 f(2x) dx$

[24] If a and b are positive numbers, show that

$$\int_0^1 x^a (1-x)^b dx = \int_0^1 x^b (1-x)^a dx$$

Exercises 6.1

[25] sketch the region enclosed by the given curves and find its area:

(a) $y = 12 - x^2$, $y = x^2 - 6$

(b) $y = \sec^2 x$, $y = 8 \cos x$, $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$

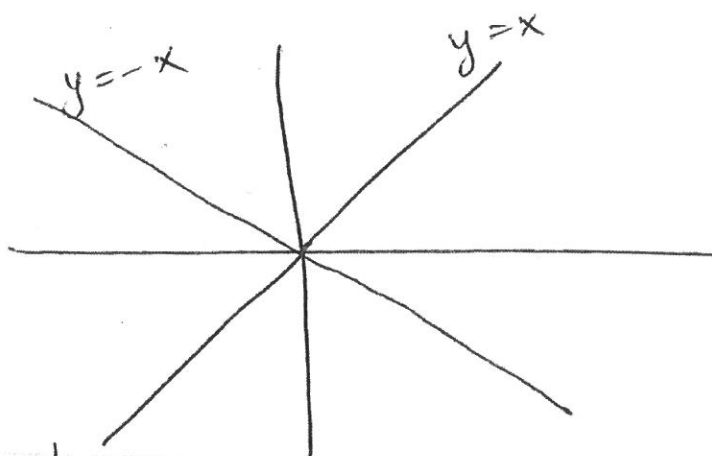
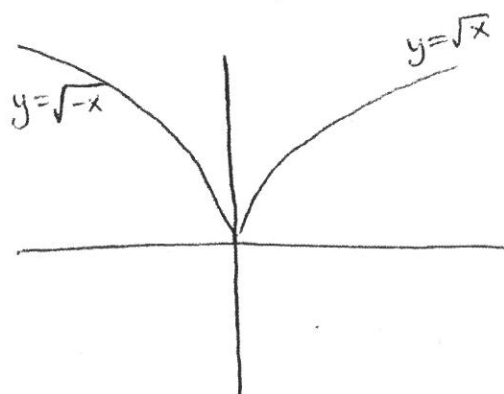
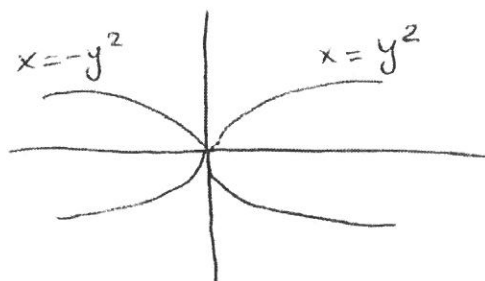
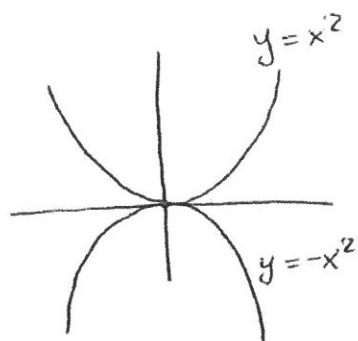
(c) $y = x^2$, $y = 4x - x^2$

(d) $y = x^3$, $y = x$

(e) $y = \sqrt{x}$, $y = \frac{1}{2}x$, $x = 9$

(f) $y = 2y^2$, $x = 4 + y^2$

Basic Curves



Shift

